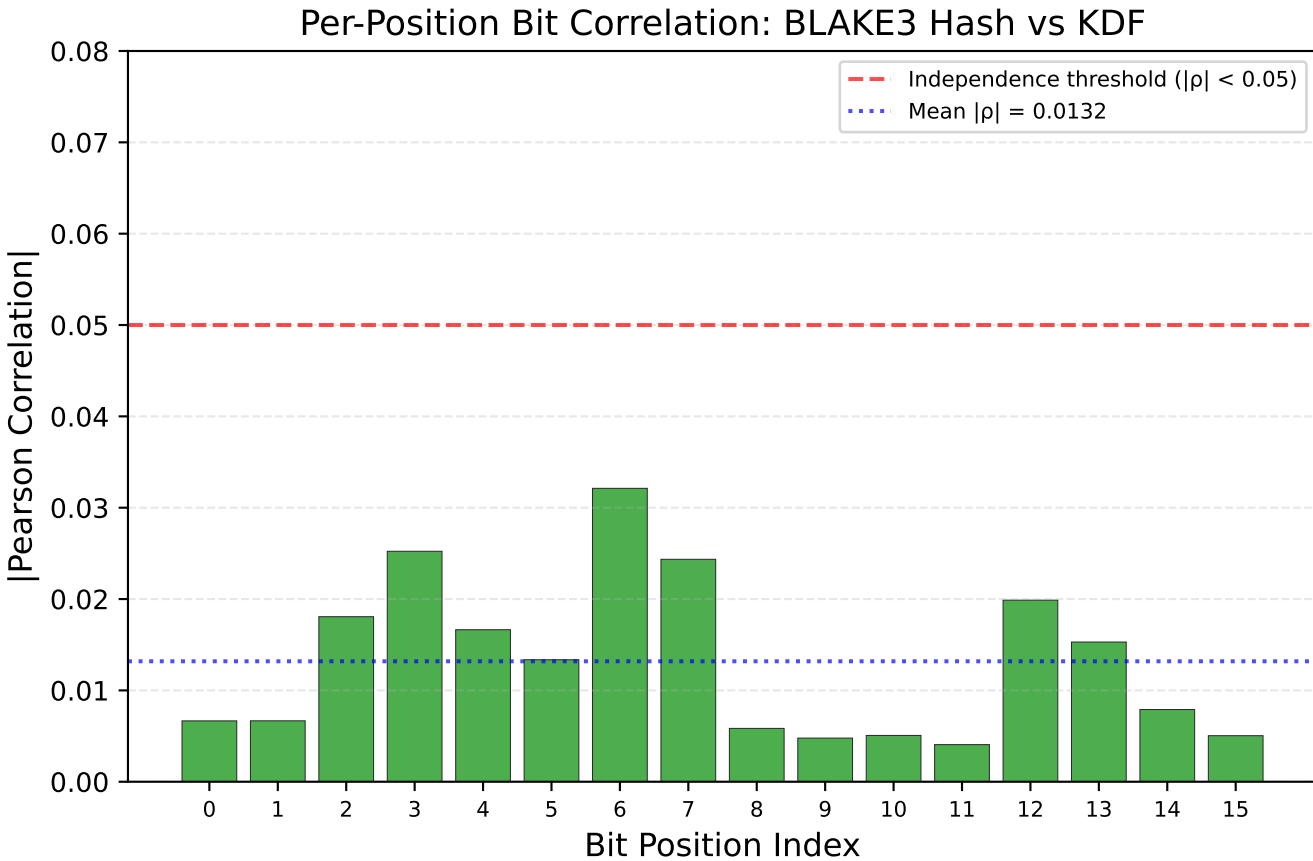


Domain Separation: Formal Independence Evidence (BLAKE3 Hash vs KDF)



Statistical Independence Test

Metric	Value	Status
Samples	3,000	—
Positions Tested	16	—
Mean Hamming Ratio	0.499943	✓ ≈ 0.5
Hamming Deviation	0.000057	✓ < 0.01
Mean ρ (all positions)	0.013186	✓
Max ρ (any position)	0.032122	✓
t-Statistic	-0.3766	—
p-Value (H₀: ρ=0)	0.7065	✓
Conclusion	No significant correlation (fail to reject H₀)	✓